

INTRODUCTION

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1.1 Introduction

Graphics

The term Graphics comes from Greek "graphikos" which means 'something written' e.g. autograph. So, Graphics are visual images or designs on some surface, such as a wall, canvas, screen, paper, or stone to inform, illustrate, or entertain.

Graphics can be created using a variety of techniques, including drawing, painting, printing, or digitally through computer software.

Whether they take the shape of graphs, diagrams, logos, or pictures, graphics often play a significant part in the visual representation of information and ideas.

Computer Graphics

- The development, modification, and presentation of visual material on a computer screen is referred to as computer graphics.
- It involves the creation and rendering of 2D and 3D visuals using software and technological tools.
- Simple line drawings and text characters on early computer systems were the initial kind of computer graphics.
- The complexity and fineness of computer graphics increased along with the capabilities of computer hardware and software.
- The area of computer graphics has expanded and diversified into a number of subfields, including computer-aided design (CAD), computer animation, virtual reality (VR), and augmented reality (AR).
- Computer graphics are employed in a variety of fields nowadays, such as video games, movies, product design, scientific simulations, and data visualization.
- The creation and modification of computer graphics is frequently done using graphics software, such as Adobe Photoshop, Autodesk 3ds Max, and Blender.

- More vivid and engaging visual experiences for users, such as immersive virtual reality worlds and realistic animated characters, are now achievable thanks to advancements in computer graphics technology.
- Graphics are a vital tool for a variety of businesses and professions since they play a crucial part in the visual representation of information and ideas.

1.2 Brief History of Computer Graphics

- **1950s:** Early computer graphics were simple line drawings created on vector displays, primarily used for scientific and engineering purposes.
- **1960s:** Ivan Sutherland creates the first interactive computer graphics program, Sketchpad, which allows users to draw and manipulate simple shapes.
- **1970s:** The first computer graphics user interfaces are developed, including the Xerox Alto and the Apple Lisa. The first video games, such as Pong, are also created.
- **1980s:** The introduction of personal computers and affordable graphics software, such as Adobe Illustrator and MacPaint, leads to the democratization of computer graphics. Pixar Animation Studios is founded, and creates the first computer-animated feature film, "Toy Story," in 1995.
- **1990s:** The World Wide Web leads to the development of web design and interactive graphics, and the introduction of 3D graphics hardware accelerators greatly improves the quality of 3D graphics. The first virtual reality systems are also developed.
- **2000s:** The rise of mobile devices and gaming leads to a renewed focus on 2D and 3D graphics optimization for performance. The use of motion capture technology in films and games becomes more prevalent.
- **2010s:** Virtual and augmented reality technologies continue to advance, creating new opportunities for immersive visual experiences. The use of artificial intelligence and machine learning in computer graphics also grows.
- **2020s:** The COVID-19 pandemic accelerates the trend towards remote work and digital communication, leading to increased demand for high-quality computer graphics in areas such as virtual events and online learning. The development of real-time ray tracing technology also improves the quality of computer graphics in video games and other applications.

In addition to traditional applications like film and gaming, computer graphics technology is also being used in new and innovative ways. For example, medical researchers are using virtual reality environments to study the human body, while architects and urban planners are using 3D visualization tools to design and plan cities.

Looking to the future, computer graphics technology is likely to continue to evolve and advance, with new technologies like augmented reality (AR) and artificial intelligence (AI) playing an increasingly important role. As these technologies continue to mature and become more widely available, we can expect to see even more exciting and innovative uses of computer graphics in the years to come.

Manual Drafting vs Computer Graphics

Feature	Manual Drafting	Computer Graphics
Creation	Drawn by hand using drafting tools such as pencils, rulers, compasses, and templates	Created digitally using software programs and input devices such as a mouse or stylus
Speed	Time-consuming, as each line must be drawn by hand	Faster and more efficient, as lines can be quickly generated and manipulated
Precision	Limited by human error and the precision of the drafting tools	Can achieve a high degree of precision and accuracy, down to the pixel level
Editability	Difficult to make changes and corrections, often requiring the entire drawing to be redrawn	Easily editable with the ability to make quick and precise changes
Reproducibility	Difficult to reproduce multiple copies with consistency and accuracy	Easily reproducible with the ability to make exact copies or modify as needed
Storage	Large physical storage space required for storing paper drawings	Digital files can be easily stored and accessed on various devices
Collaboration	Difficult to collaborate with others on the same drawing, as physical copies must be shared	Can easily collaborate with others in real-time, even remotely

Feature	Manual Drafting	Computer Graphics
Cost	Low cost in terms of equipment, but high cost in terms of time and labor	The high initial cost for software and hardware, but the lower ongoing cost for editing and reproduction

Overall, while manual drafting is still used in certain fields, computer graphics has largely replaced it in many applications due to its speed, precision, editability, and reproducibility.

Characteristics of Computer Graphics

1. **Precise:** Computer graphics have the potential to be extremely exact, enabling accurate depiction of both data and visuals.
2. **Flexibility:** The creation and modification of visuals are more flexible when using software and digital technologies.
3. **Interactivity:** Interactive computer graphics enable real-time user manipulation and interaction with visual data.
4. **Realism:** The development of computer graphics technology has allowed for the production of extremely realistic and lifelike images and animations.
5. **Speed:** Since computer graphics can be promptly made and changed, it is feasible to produce numerous iterations of a graphic or make adjustments on the fly.
6. **Scalability:** Images may be enlarged or shrunk without reducing the quality or resolution.
7. **Reproducibility:** Digital graphics are easily reproduced, making it simple to make several copies or repetitions of a picture.
8. **Integration:** Digital media, including websites, video games, and movies, may easily include computer visuals.
9. **Multidimensionality:** Computer graphics may be made in two or three dimensions (2D or 3D), providing more depth and complexity in visual representations.
10. **Consistency:** Computer graphics have the ability to remain constant in terms of color, layout, and other design features, allowing for the maintenance of a consistent visual identity throughout many graphics and projects.

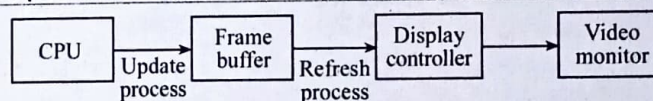
Classification of computer graphics

1. **2D and 3D Graphics:** Graphics can be classified into two-dimensional (2D) and three-dimensional (3D) graphics. 2D graphics are flat, using only the X and Y axes to represent images,

while 3D graphics add the Z axis, which allows for more depth and realism.

2. **Vector Graphics and Raster Graphics:** Vector graphics are created using geometric shapes such as lines, curves, and polygons. These images can be scaled without losing quality. Raster graphics, also known as bitmap graphics, are made up of a grid of pixels and can become pixelated when scaled.
3. **Interactive and Non-Interactive Graphics:** Interactive graphics are those that allow the user to interact with the content, such as in a video game or virtual reality environment. Non-interactive graphics, such as those used in films or advertisements, are passive and do not respond to user input.
4. **Real-Time and Non-Real-Time Graphics:** Real-time graphics are generated on the fly and displayed in real-time, such as in video games or simulations. Non-real-time graphics, such as those used in film or pre-rendered animations, are created beforehand and played back at a later time.
5. **2D Animation and 3D Animation:** 2D animation involves the creation of a sequence of images that are played back in sequence to give the illusion of motion. 3D animation involves the creation of animated characters or objects using three-dimensional models that can be manipulated in real-time.
6. **Scientific Visualization and Information Visualization:** Scientific visualization is used to represent data from scientific or engineering applications, while information visualization is used to represent complex data sets such as financial data or social networks.
7. **Digital Imaging and Photo-Realistic Rendering:** Digital imaging involves the manipulation of photographs or other digital images, while photo-realistic rendering involves the creation of highly realistic 3D images that are indistinguishable from photographs.

Components of Computer Graphics



Digital memory buffer:

- It is part of a computer's graphics hardware that stores digital images and video frames.
- It holds the digital representation of an image or video frame before it is displayed on a monitor or output device.

- The digital memory buffer allows the computer to manipulate and process the image or video frame before displaying it on the monitor.
- It can be accessed very quickly by the computer's graphics hardware.
- It is usually located on the graphics card or integrated into the motherboard.
- The size of the digital memory buffer determines the maximum resolution and color depth of the displayed images.

TV monitor:

- A TV monitor is a type of display device that receives and displays broadcast television signals.
- It can also be used to display computer-generated images and video.
- It typically uses cathode ray tube (CRT) technology or liquid crystal display (LCD) technology.
- CRT monitors are bulky and have lower resolution than LCD monitors, but they can display more accurate colors.
- LCD monitors are thin and lightweight, have higher resolution, and use less power than CRT monitors.
- The size and resolution of a TV monitor determine the quality and clarity of the displayed images.

Display controller:

- A display controller is a component of a computer's graphics hardware that manages the communication between the computer and the display device.
- It sends digital signals to the display device to create images or video.
- It is responsible for controlling the refresh rate, resolution, color depth, and other display settings.
- It can be integrated into the graphics card or integrated into the motherboard.
- It can support various types of display devices, such as CRT monitors, LCD monitors, and projectors.
- The quality and performance of a display controller affect the quality and performance of the displayed images or video.

1.3 Application of Computer Graphics

- Computer-aided design (CAD) software for engineering and architecture
- Gaming and entertainment industry applications
- Film and television production software
- Graphic design and digital art creation software
- Image and video editing software
- Data visualization and business intelligence software
- Geographic information systems (GIS) software
- Medical imaging and diagnostic software
- Scientific visualization software
- Product packaging and prototyping software
- Advertising and marketing materials design software
- Interactive maps and mapping software
- Web and mobile application user interface (UI) design software
- Industrial and product design software
- Textile design and printing software
- Animation and special effects software for movies and TV
- Video game engine software
- E-learning software and courseware development tools
- Visual simulations for training and education
- 3D printing and rapid prototyping software.
- Internet

Some of them are described below

1. Graphic design:

- Create visually appealing designs for a variety of purposes including branding, advertising, and marketing.
- Manipulate images, create vector graphics, and edit text using software such as Adobe Photoshop, Illustrator, and InDesign.
- Computer graphics allow the designer to create high-quality images, logos, and other graphics for use in print media, web design, and other digital marketing campaigns. This application of computer graphics plays an important role in the visual communication industry.

2. Video game development:

- Production of game content, characters, and environments. - Create 3D models, textures, animations, and environments using software such as Autodesk Maya, Unity and Unreal Engine.
- Computer graphics play an important role in the development of video games, allowing developers to create immersive game environments, realistic characters and detailed 3D models of him that can be used on various game development platforms.

3. Film/animation production:

- Create special effects, 3D models and animation sequences for movies and animations.
- Create realistic animations, simulations, and visual effects using software such as Autodesk Maya, Cinema 4D, and Houdini.
- Computer graphics play a key role in film and animation production, allowing animators and special effects artists to create characters and environments that come to life and add visual effects not possible in the real world.

4. Architectural Visualization:

- Creation of his 3D models of buildings and landscapes.
- Realistic visualization of proposed designs using software such as SketchUp, Autodesk Revit and Lumion.
- Computer graphics are important in visualizing architectural designs by creating high-quality 3D models that can be rendered and viewed from different angles to provide a realistic view of the proposed design. play a role.

5. Industrial design:

- Production of product prototypes and models.
- Create his 3D model of the product using software such as SolidWorks, CATIA, AutoCAD.
- Computer graphics play a key role in the industrial design process, allowing designers to create detailed 3D models of products to virtually test before creating physical prototypes.

6. Medical Imaging:

- Diagnosis of diseases through visualization and analysis of medical data.
- Using software such as MRI and CT scanners to generate volumetric data that can be rendered using computer graphics software to create his 3D visualization of organs and tissues.

- Computer graphics play an important role in medical imaging as they enable doctors and researchers to visualize and analyze complex medical data to diagnose and treat medical conditions.

7. Virtual Reality:

- Create immersive environments and simulations.
- Create virtual reality environments using software such as Unity and Unreal Engine.
- Computer graphics play a key role in virtual reality, creating high-quality 3D models and environments that can be rendered in real-time to provide an immersive experience for users.

8. Augmented Reality:

- Superposition of digital information and the real world.
- Create augmented reality applications using software such as ARKit and ARCore.
- Computer graphics play a key role in augmented reality, overlaying digital information on top of the real world to provide users with an interactive experience.

9. Scientific visualization:

- Creating visual representations of complex data and scientific concepts.
- Visualize scientific data using software such as MATLAB, VisIt, and ParaView.
- Computer graphics play a key role in scientific visualization, creating high-quality visual representations of complex data and scientific concepts that can be used for research and educational purposes.

10. Education and Training:

- Create interactive learning experiences for students and professionals.
- Create e-learning courses using software such as Adobe Captivate, Articulate Storyline, and Lectora.

11. Cartography:

Computer graphics is used in cartography to create maps and visual representations of geographical data.

12. Fashion Design:

Computer graphics is used in fashion design to create digital models and visualizations of clothing designs.

13. Digital Art:
Computer graphics is used in digital art to create original pieces of art using digital tools and techniques.

14. Sports Graphics:
Computer graphics is used in sports broadcasting to create visualizations of game statistics, replays, and highlights.

3 broad Elements Of computer Graphics

1. Geometric modelling

- Mathematically expressing physical things and their attributes
- Building 3D representations of objects using points, lines, curves, and surfaces
- Used in a variety of applications, including CAD, animation, and virtual reality

2. Rendering:

- Creating a 2D picture or animation from a 3D model or scene
- Calculating an object's color, shading, and texture depending on its position and illumination
- Used to produce photorealistic visuals and animations for movies, video games, and advertising

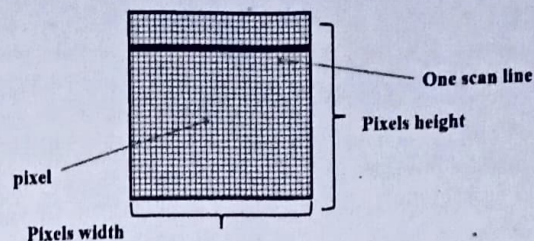
3. Display:

- Showing a computer graphics application's output on a screen or other output device.
- Display technology comprises monitors, projectors, and printers. -Converting digital data into a format that can be shown or printed.
- Users can engage with and control computer graphics programs thanks to the user interface.

Some Important Terminology in Computer Graphics

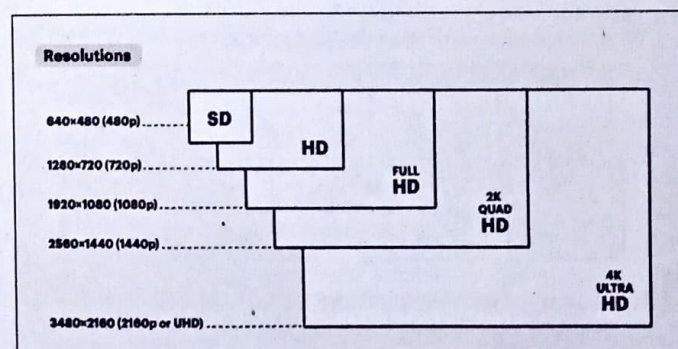
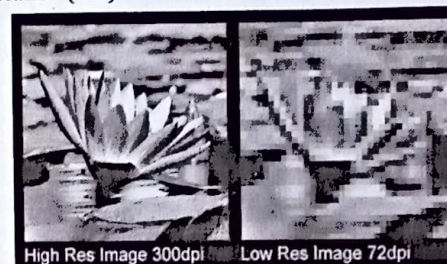
1. Pixel:

- Pixels are the smallest unit of an image and are frequently used to quantify resolution.
- A pixel is a single point in a digital picture.
- Picture element is where the word "pixel" originates.
- Each pixel has its own gray level or intensity name or address by which we can control.



2. Resolution:

- The number of pixels in an image is its resolution.
- The number of pixels contained on a display monitor, expressed in terms of the number of pixels on the horizontal axis and the number on the vertical axis.
- It is often calculated as width $\times$ height (e.g., 1920×1080).
- Photographs with a higher resolution will be more detailed.
- The sharpness of the image on a display depends on the resolution and the size of the monitor.
- Pixels per inch are a common unit used to describe display resolution (PPI).



3. PPI

- PPI stands for "Pixels Per Inch".
- It measures the resolution of a display or image.
- PPI indicates how many pixels are present in a linear inch of the display or image.
- A higher PPI results in a higher level of detail and clarity in the image or display.
- PPI is commonly used to describe the resolution of displays such as computer monitors, smartphones, and televisions, as well as the resolution of digital images.
- A higher PPI is generally considered desirable for high-quality displays and images, as it results in a sharper and more detailed image.
- However, a higher PPI also requires more processing power and memory to display and manipulate the image, which can impact performance.

4. Aspect Ratio

- The proportional relationship between the width and height of an image or display is referred to as the aspect ratio.
- The width is always listed first and is expressed as the width-to-height ratio.
- Some aspect ratios are better suited for particular types of content than others, and they are used for various purposes.
- Because it offers a more immersive viewing experience, widescreen aspect ratios like 16:9 are frequently used for movies and television shows.
- Because it enables the subject of the image to be centered and easily visible, a square aspect ratio like 1:1 is frequently used for social media images and profile pictures.

5:4 Computer Displays	4:3 SDTV / Video Computer Displays	3:2 35mm Film DSLR Cameras Smartphones	16:10 Widescreen Computer Displays Smartphones
16:9 HDTV Widescreen HDTV Smartphones	1.85:1 Cinema Film (US)	2.35:1 Cinemascope	

What is the difference between 16:9 and 4:3

- The 16:9 aspect ratio is 78% wider than it is tall.

- In comparison, the 4:3 aspect ratio is 33% wider than tall. Thus, the 16:9 aspect ratio can fit more information horizontally, while the 4:3 aspect ratio has more space vertically.

Mathematically, as the relationship between the width and the height is shown in the ratio of 4:3, if one divides the width by the height is:

$$4 \div 3 = 1.33$$

This makes it 1.33:1, which serves as the aspect ratio of a 4:3 TV or projector screen. However, this aspect ratio only remained as long as the shape of the television remained square.

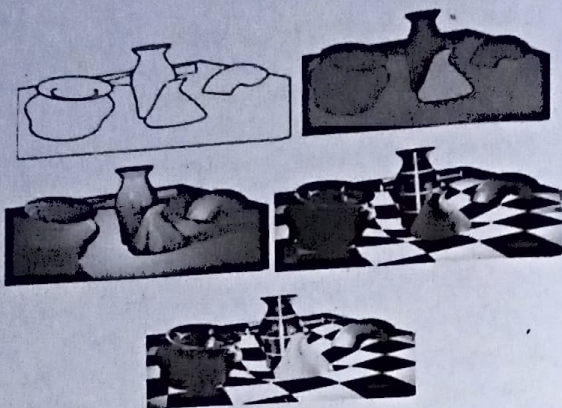
Now, the 16:9 aspect ratio becomes 16 units wide and 9 units high- much wider in comparison to the 4:3 aspect ratio. Also, when the width is divided by the height it is:

$$16 \div 9 = 1.777$$

This makes it 1.78:1, which is a perfect fit for wider-sized televisions.

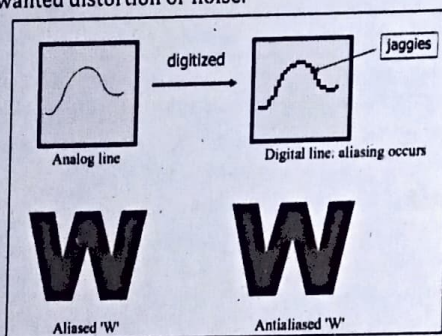
5. Rendering

- Rendering, also known as image synthesis, is the process of using a computer program to create a photorealistic or non-photorealistic image from a 2D or 3D model.
- To simulate the appearance of real-world objects and environments in a virtual environment is the purpose of rendering.
- A scene is given lighting, shading, and other visual effects, and then a 2D image-or series of images-that depicts the scene from multiple angles is created.
- The modeling, texturing, lighting, and compositing stages are among the stages that make up the rendering process.
- Modeling entails generating 3D models of real-world or manufactured objects using specialized software.
- Texturing effectively gives the 3D models surface details like colors, patterns, and textures.
- Lighting simulates the effects of light, such as shadows, reflections, and other effects, on the objects in the scene.
- Compositing is the process of combining the various scene components, such as backgrounds, foregrounds, and special effects, to create the final image or animation.
- Rendering is used in many different fields, such as video game development, architecture, product design, and visualization.



6. Aliasing

- When a digital image or signal is sampled or displayed at a low enough resolution or frequency, aliasing—a distortion or jaggedness—occurs in the data.
- The term "jaggies" refers to the appearance of jagged or pixelated edges of objects in computer graphics when aliasing is present on a screen with a low pixel density.
- By softening the jagged edges of the image's objects, anti-aliasing techniques can be used to minimize the appearance of aliasing.
- A higher sampling rate or the use of filtering techniques to eliminate frequencies above the Nyquist frequency can minimize aliasing, which can also occur in audio signals and cause unwanted distortion or noise.



7. Bit depth (Color depth)

- The term "bit depth" describes the quantity of bits used to express each pixel's color in a digital image.

- The range of colors that can be represented increases with the number of bits used for each pixel, and the image appears more accurate and detailed.
- Typically, bits per pixel (bpp) or bits per channel are used to express bit depth (bpc).
- Digital images frequently use bit depths of 8-bit (256 colors), 16-bit (65,536), and 24-bit (16.7 million colors).
- In order to preserve the quality and fidelity of the image during editing and manipulation, high bit depth is frequently used in professional graphics applications like photo editing and color grading.

8. Interlacing

- Interlacing is a technique for splitting each frame into two fields to display images on a screen.
- The image's even-numbered lines are all contained in the second field, while all the odd-numbered lines are contained in the first field.
- The image appears to be displayed in one go because the fields are alternately displayed on the screen.
- On CRT monitors, interlacing can lessen flicker, but it can also result in distorted images and other issues.
- Nowadays, progressive scanning on modern displays, which shows the entire image at once, has rendered interlacing largely obsolete.

Interlaced Scan



9. Persistence

- A pixel on a display remains lit for a certain amount of time after it has been turned on, which is known as persistence.
- The persistence of CRT displays is determined by how long their phosphor coatings glow after being struck by an electron beam.
- Though it can lessen flicker and enhance the look of moving images, higher persistence can also result in motion blur and decreased sharpness.

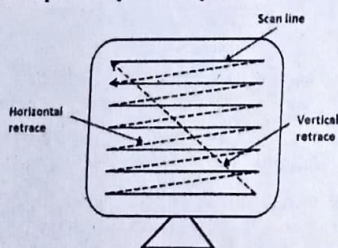
- Less blurring and faster response times are indicated by persistence values that are typically expressed in milliseconds.
- Given that modern LCD and OLED displays employ various illumination techniques without persistence, persistence is less of an issue with these displays.

10. Refresh rate

- The number of times per second that an image is redrawn on a display is referred to as the refresh rate.
- A faster refresh rate can lessen flicker, enhance the clarity of moving objects, and ease eye strain and fatigue.
- Higher numbers denote a faster refresh rate, which is typically measured in Hertz (Hz).
- Displays frequently have refresh rates of 60 Hz, 120 Hz, and 144 Hz.
- The display's quality, the graphics card, and the connection between the two can all have an impact on the refresh rate.
- A refresh rate of 75 HZ means that the image is redrawn 75 times a sec.

11. Horizontal and vertical retrace

- The terms "horizontal retrace" and "vertical retrace" describe the times that the electron beam in a CRT display needs to travel from one line's end to the next's beginning and from one frame's end to the next's beginning, respectively.
- These times result in the display briefly going dark as the electron beam is stopped and returns to its initial position.
- To guarantee that the electron beam begins in the proper location for the following line or frame, retrace periods are required.
- The flicker and stability of the display, as well as the timing and synchronization of the display with other system components, can all be impacted by retrace periods.



12. Refresh buffer

- A section of computer memory used to store images temporarily before they are displayed on screens.
- In order to keep the image on the screen current, it is updated frequently.
- Also known as a screen buffer, frame buffer, or video buffer.
- It is in charge of storing the bits that signify each pixel's color and intensity.
- The graphics hardware typically accesses it directly from video memory where it is typically stored.

13. Bit plane

- A method for storing images in computer memory that treats each bit of a byte as a separate plane.
- The first plane is the least detailed, and the last plane is the most detailed, with each plane denoting a different level of detail.
- In computer graphics, it is employed for texture mapping, image compression, and special effects.
- Additionally, it is employed for the display of grayscale images, such as those found in the medical field, where each bit plane designates a different shade of gray.
- A full-color image can be produced by combining bit planes.

14. Bit map

- A 2D grid of pixels, where each pixel corresponds to a discrete region of the image.
- A color and intensity value is given to each pixel.
- In computer graphics, it is a typical method of representing digital images.
- It is also known as a raster image or pixel map.
- Bitmap images can be saved in a variety of file types, including BMP, JPEG, and PNG.

15. Fluorescence and phosphorescence

- The process of emitting light when a material is excited by absorbing light of a particular wavelength is known as fluorescence.
- Similar to fluorescence, phosphorescence emits light that lasts long after the excitation source is removed.

- In computer graphics, both phenomena are used to produce special effects like glowing objects and vivid colors.
- They are also utilized in display technologies like fluorescent lights and cathode ray tubes (CRTs).
- Computer graphics can simulate fluorescence and phosphorescence by changing the hue and brightness of the light that various elements of a scene emit.

Display size of Mobile

Your mobile phone has a total resolution of 1920×1080 with 423 PPI. What is the display size of your mobile?

Here, resolution = 1920×1080

PPI = 423 PPI

Now,

Using Pythagoras theorem

$$X^2 = 1920^2 + 1080^2 = 4852800$$

$$X = 2202.9071$$

$$\text{PPI} = \frac{\text{Resolution diagonally}}{\text{Screen size in inch}}$$

On solving, Screen size = 5.2 inch

